

Web Introduction

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World Wide Web (WWW)

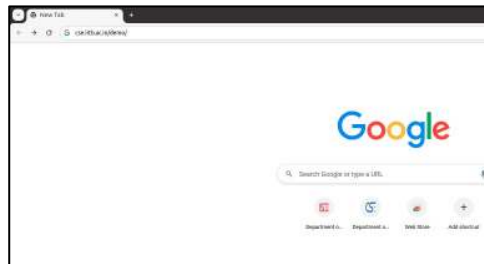
- A service provided over Internet, enabling users to access and interact with a vast array of information and services

Terminology

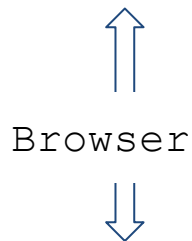
- Web Server: a software application hosted on a machine, that delivers web content/service
 - Browser requests arrive here
 - Serves **static** content
 - Examples: Apache, Nginx, Microsoft IIS
- Web Client (Browser): Application running on user's device
 - Sends out requests to server, processes received responses and renders web pages

- Web Application: a software program or set of programs that run on a machine
 - Help **dynamically** generates content in response to requests
 - Involve server-side scripting, database interactions, and complex business logic
 - E.g. Amazon, Facebook, Google docs, BodhiTree etc

Time



HTTP Request

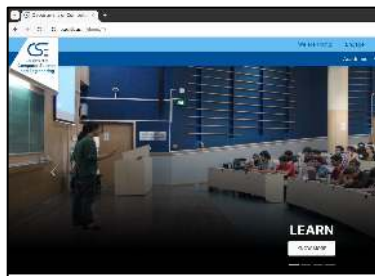


WWW

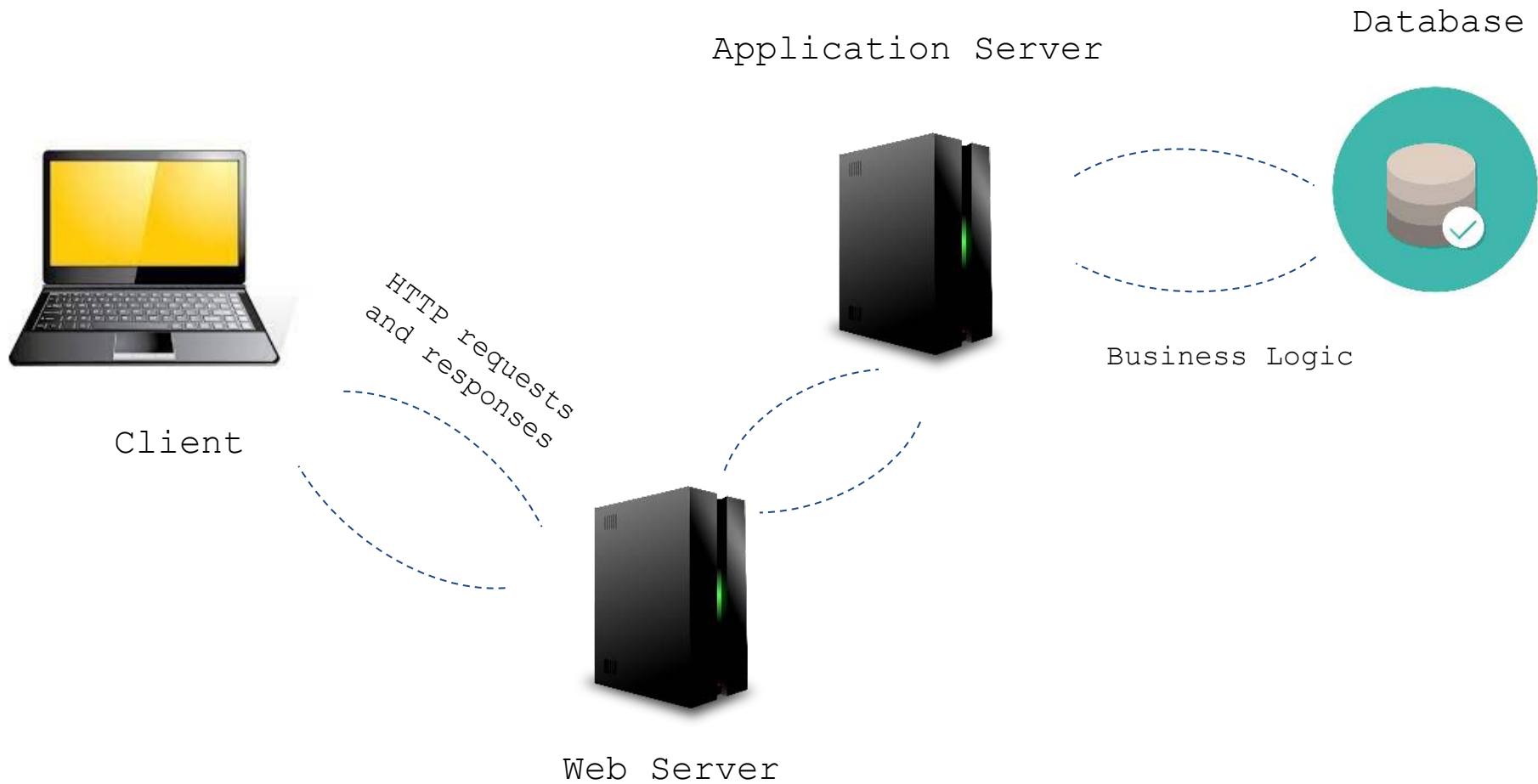


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Web Server



HTTP Response



Web Client-Server Interaction

Typical Workflow (At high level)

- Web Request:
 - A user enters a web address (URL) in browser
 - One or more requests sent to web server hosting website
- Processing:
 - Server/Application processes each request
 - May run server-side scripts, query databases, or performing other tasks to generate a response

- Response:
 - The generated response(s) sent back to user's browser
- Delivery:
 - Browser interprets response(s) and displays web page
- Repeat above steps:
 - User interacts with the page causing more requests to be sent out
 - Receives corresponding responses and sees new information, and so on

Outline

- What constitutes a webpage?
- What goes on inside a Browser (web client)?
 - What standard security mechanisms implemented?
- How do client and server communicate?
 - HTTP/HTTPS protocol
 - Session Management via cookies and tokens
- How does a web server process requests and generate responses?
 - Static vs Dynamic content